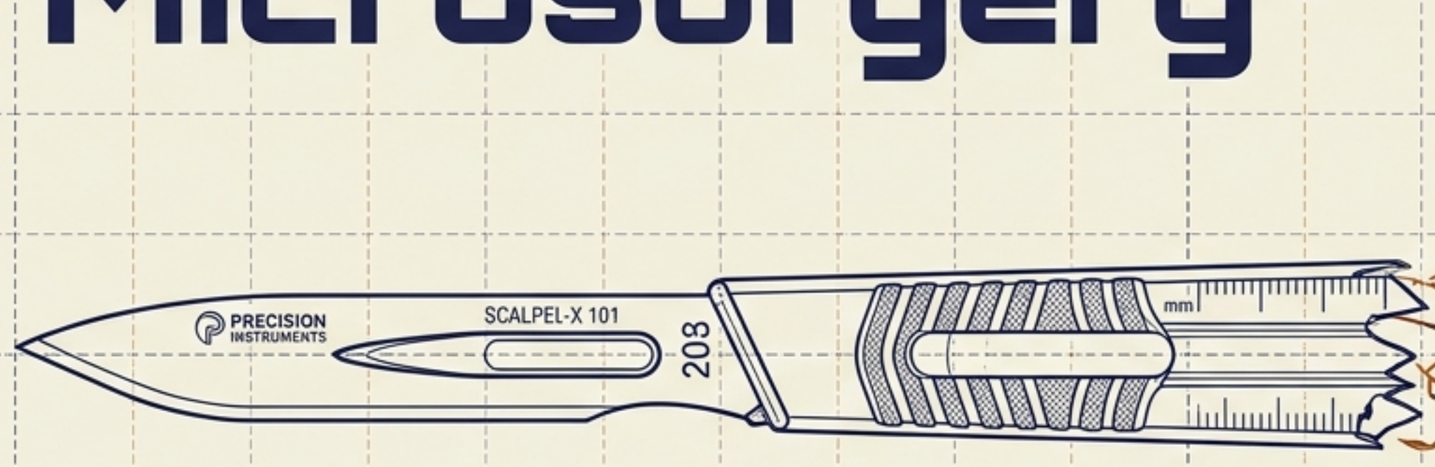
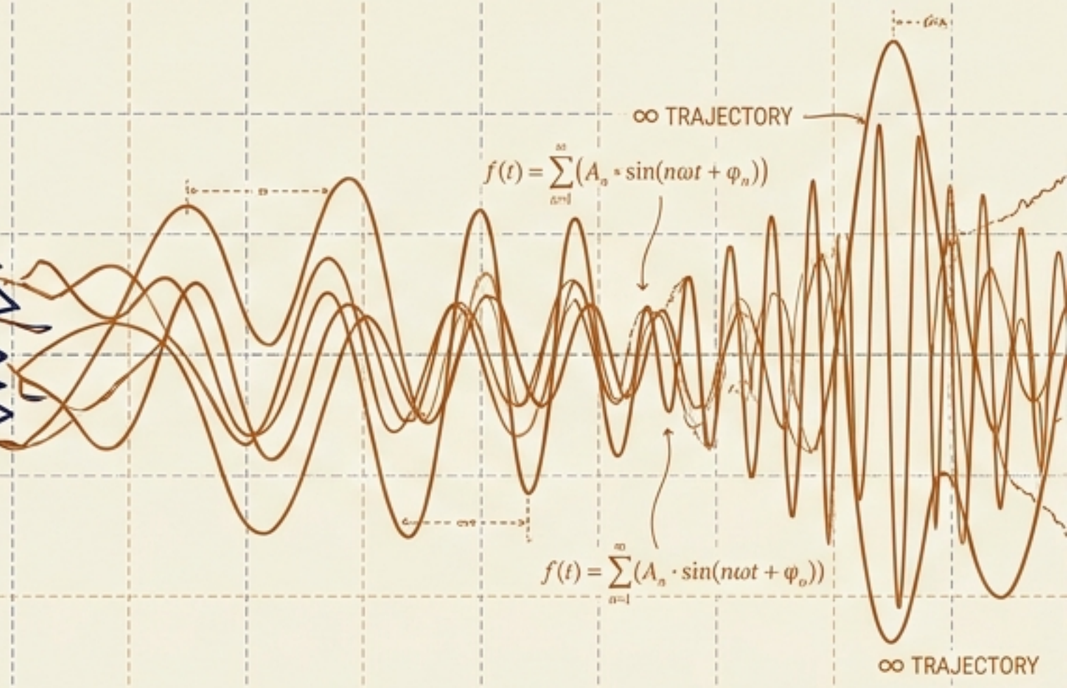
Continuous Microsurgery



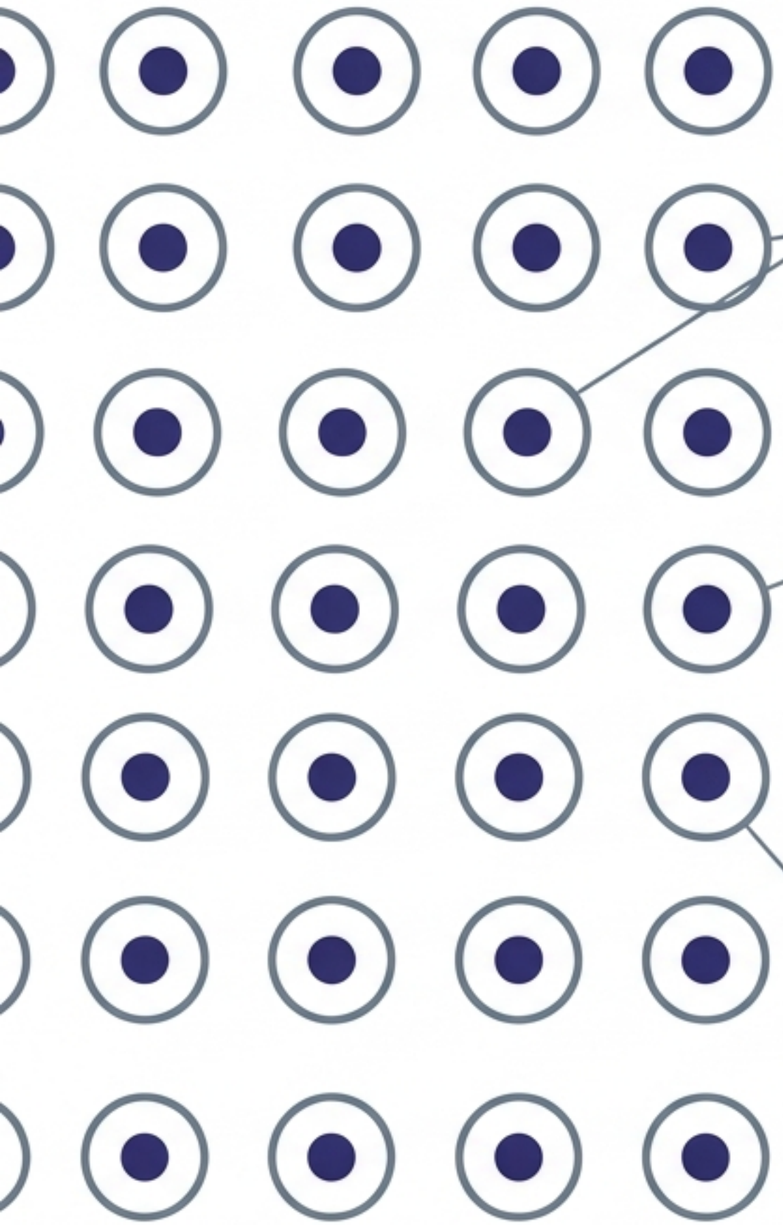


The Ethics of Never Putting the Knife Down

Why longitudinal medicine requires a new unit of ethical analysis.

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The Episodic Lens (Status Quo)

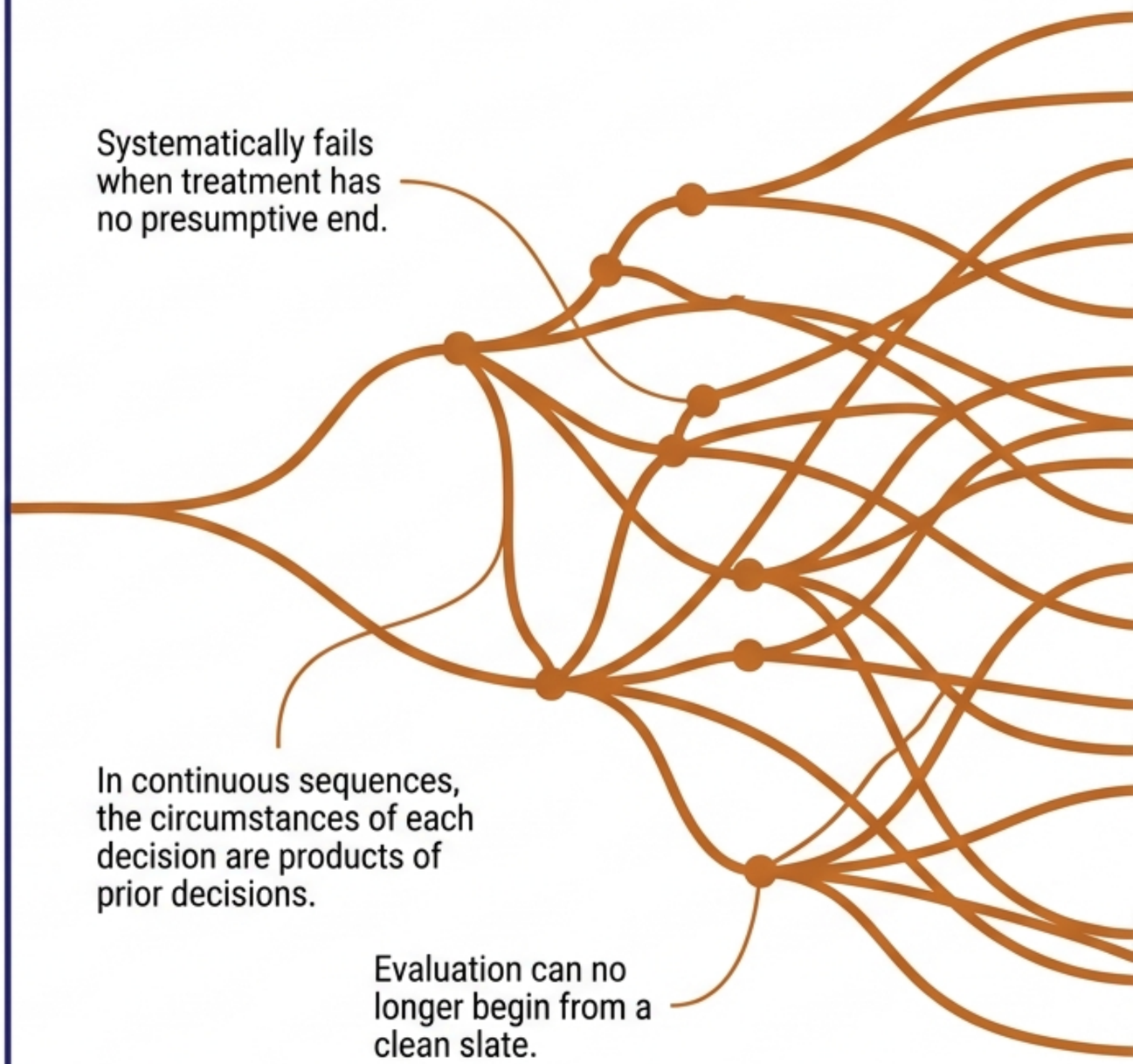


Evaluates interventions one at a time.

Focuses on immediate indication, consent, and expected harm vs. benefit.

Perfectly suited for episodic medicine (e.g., operations, antibiotics) with a distinct beginning and end.

The Trajectory Lens (The Reality)



Systematically fails when treatment has no presumptive end.

In continuous sequences, the circumstances of each decision are products of prior decisions.

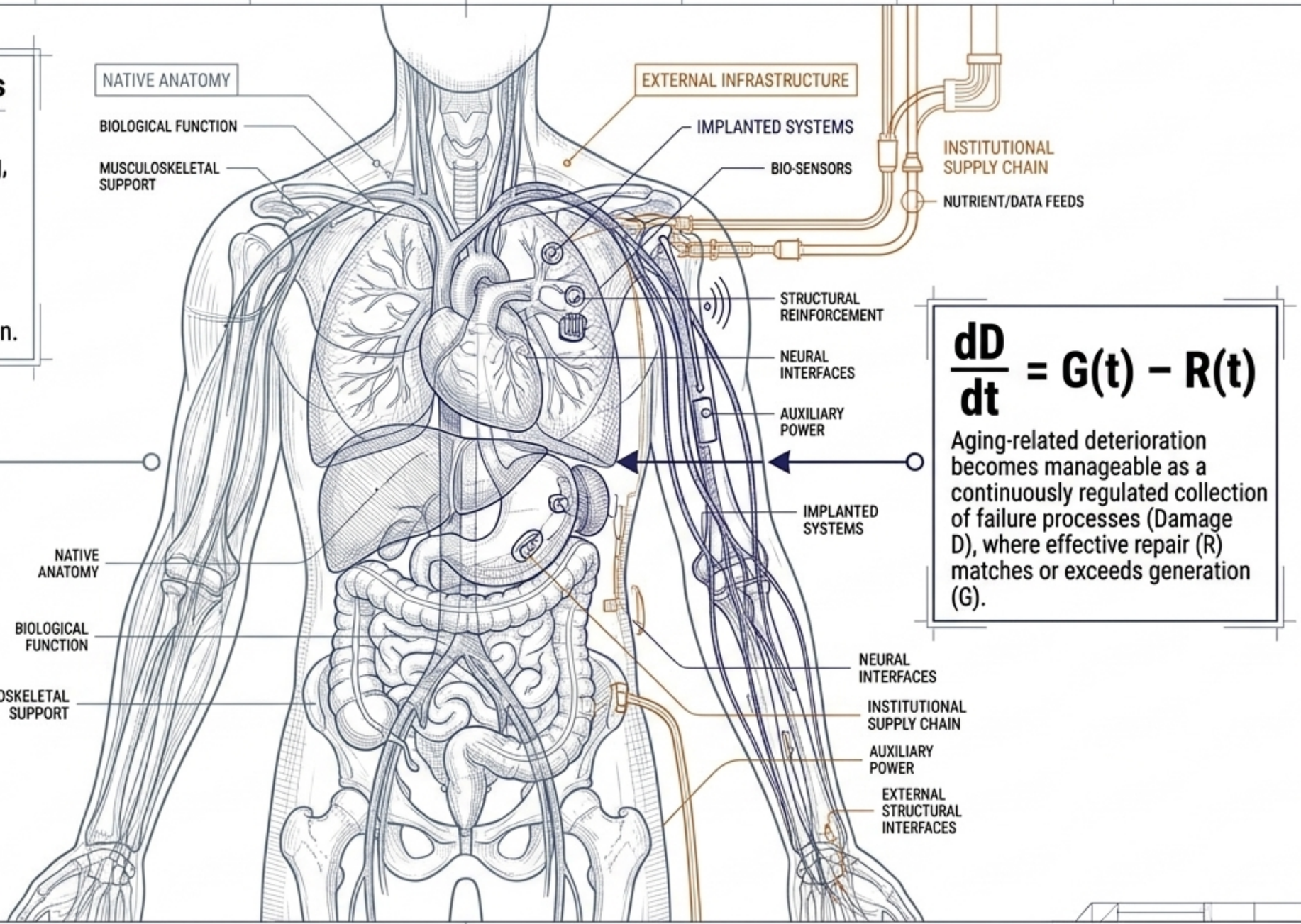
Evaluation can no longer begin from a clean slate.

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The Longevity Hypothesis

Extending healthy life not through a single cure for aging, but by progressively replacing the body's dependence on its original anatomical configuration.

Reconstruction over Preservation.



$$\frac{dD}{dt} = G(t) - R(t)$$

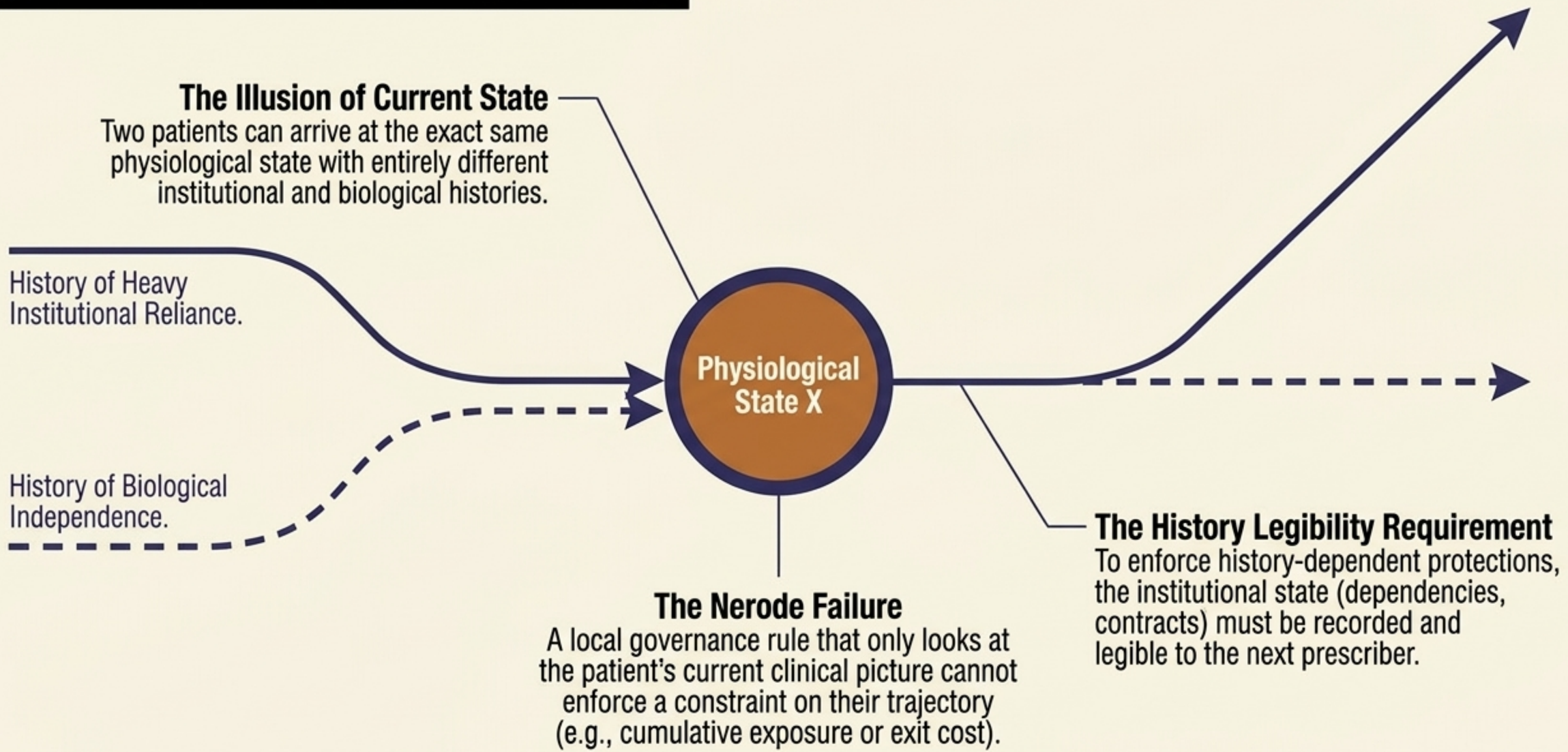
Aging-related deterioration becomes manageable as a continuously regulated collection of failure processes (Damage D), where effective repair (R) matches or exceeds generation (G).

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System	Longevity Function	Evidential Status	The Quantitative Constraint
Hepastitium (Redundancy)	Distributed sensing and auxiliary regulation.	Hypothetical	Redundancy only extends unmaintained viability logarithmically; strictly bounded by common-cause failure.
Umbilicus (Metabolism)	External nutrient and waste exchange.	Hypothetical (infection/thrombosis risks)	Mass balance trap: Light use can worsen physiological balance by accelerating native atrophy.
Skeletal Adjustment	Gradual structural adaptation (0.5mm / 5hrs).	Partly established (distraction osteogenesis)	Asymmetric return times: Tissue strain limits dictate that meaningful reversal takes much longer than advancement.
Micro-repair Bots	Autonomous tissue maintenance.	Speculative	The Damage Floor: A mathematical threshold below which bots introduce more net harm than the damage they remove.

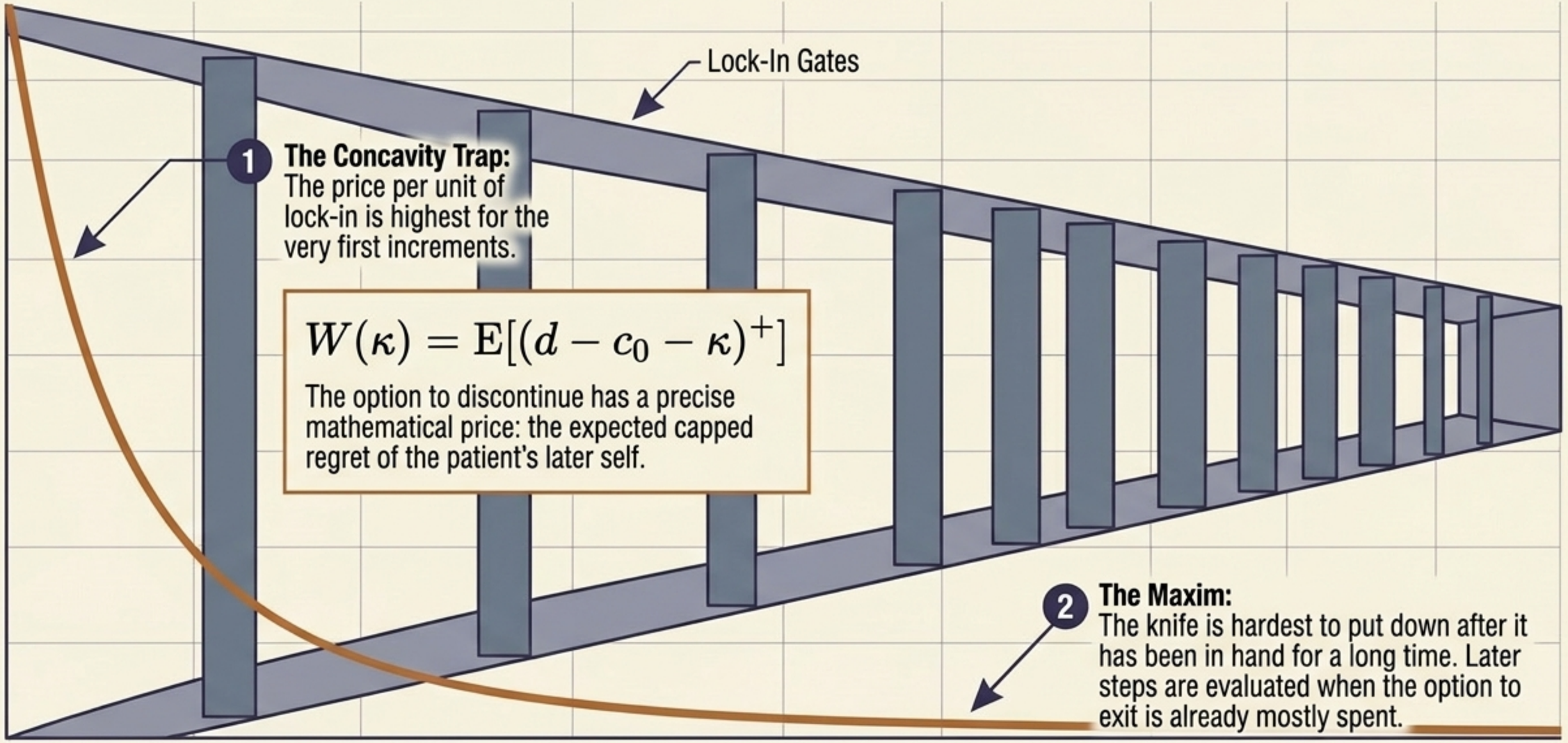
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The Nerode Failure in Patient Trajectories



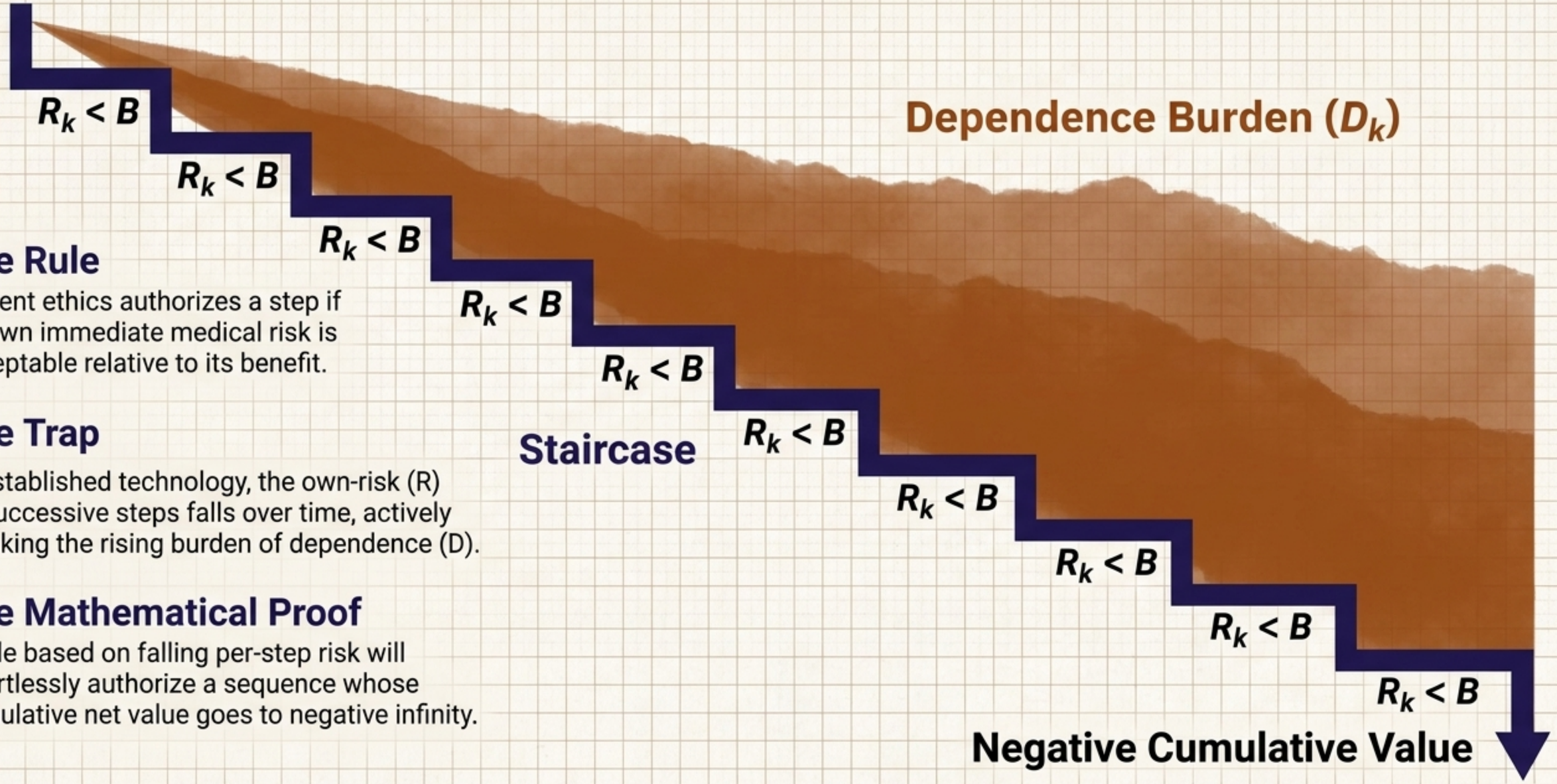
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Pricing the Option to Discontinue



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The Greedy Authorization Trap



The Rule

Current ethics authorizes a step if its own immediate medical risk is acceptable relative to its benefit.

The Trap

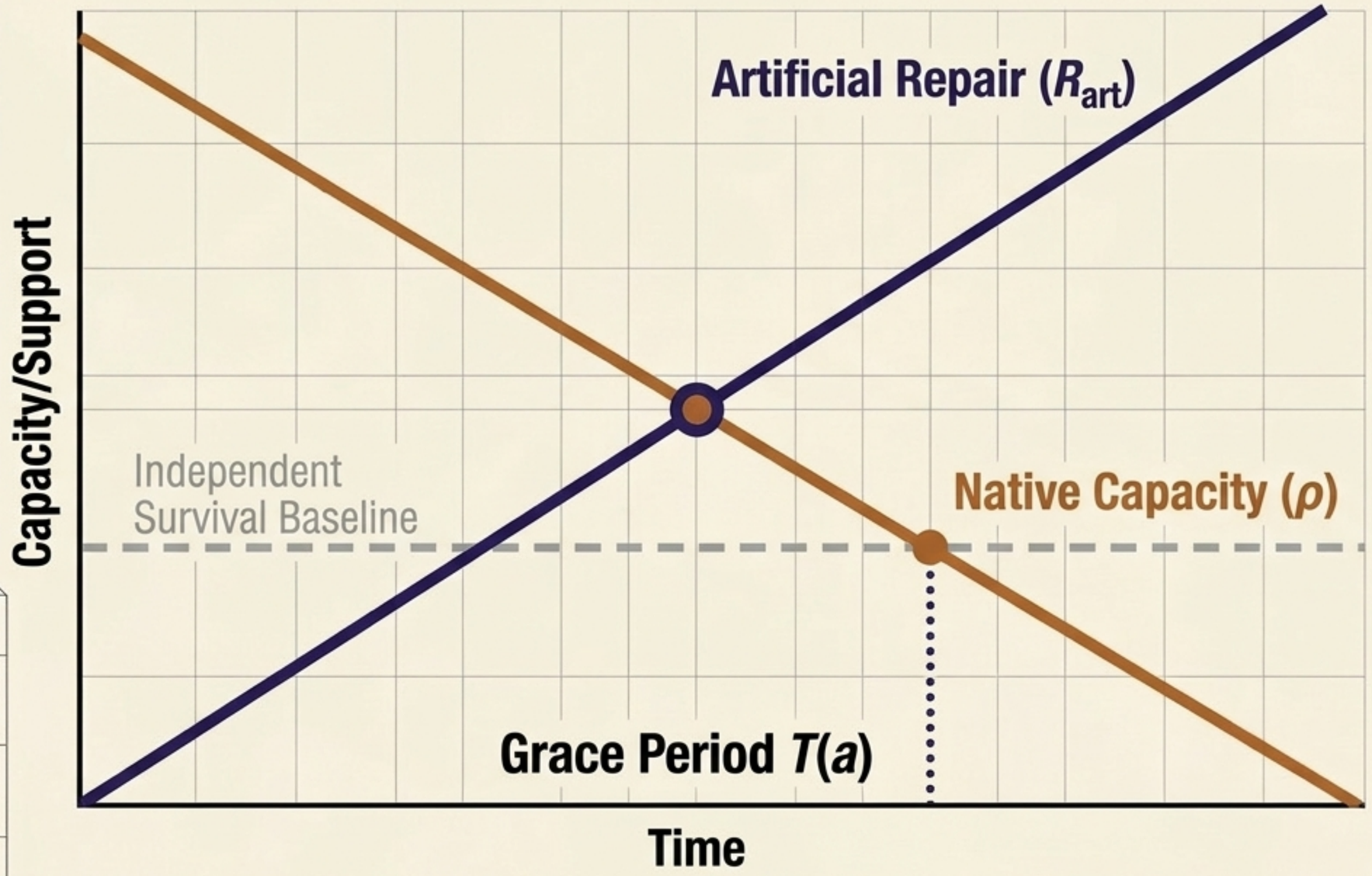
In established technology, the own-risk (R) of successive steps falls over time, actively masking the rising burden of dependence (D).

The Mathematical Proof

A rule based on falling per-step risk will effortlessly authorize a sequence whose cumulative net value goes to negative infinity.

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The Biological Ratchet (The Atrophy Trap)



The Mechanism:

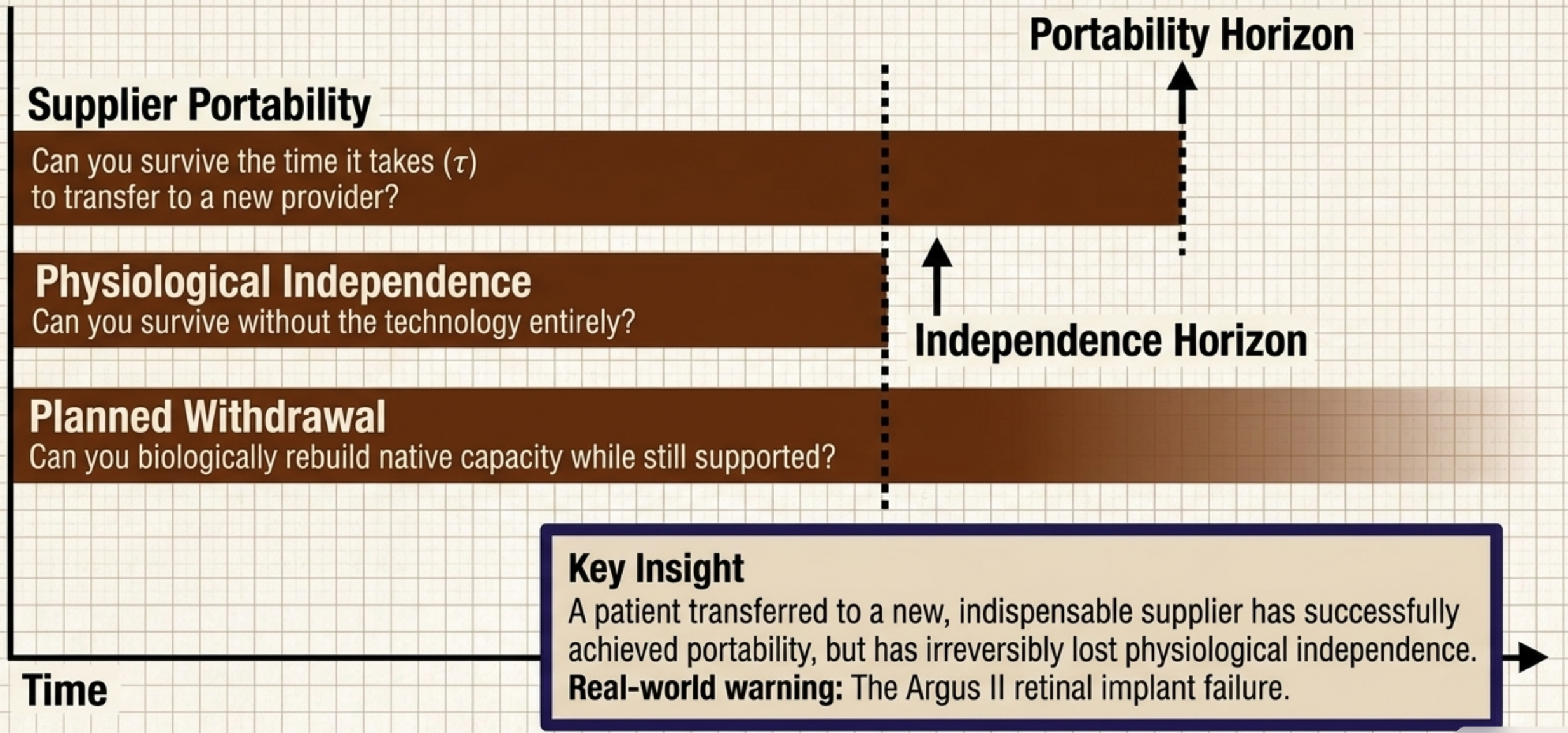
The more the regime repairs, the less the body does. The less the body does, the more the regime MUST repair.

The Consequence:

Use-dependent atrophy continuously shortens the Grace Period, turning a functional dependency into an immediate emergency of survival.

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The Exit Matrix: Portability, Independence, & Withdrawal

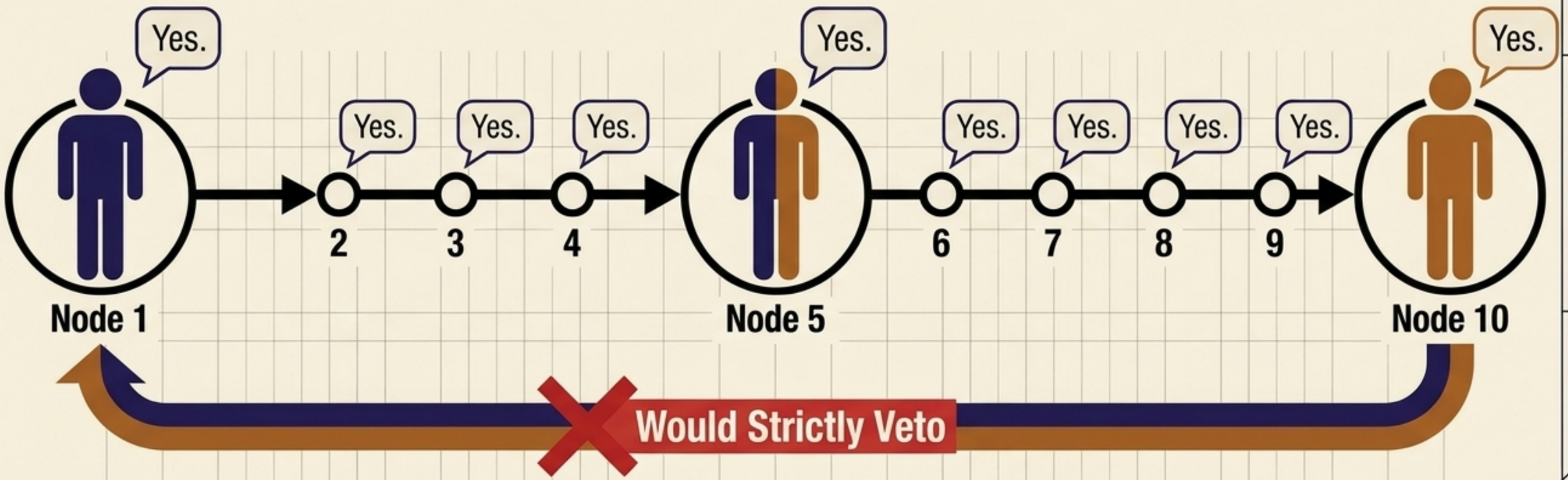


Key Insight

A patient transferred to a new, indispensable supplier has successfully achieved portability, but has irreversibly lost physiological independence.
Real-world warning: The Argus II retinal implant failure.

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The Illusion of Stepwise Consent



Endogenous Preferences

The interventions themselves alter what the person wants and what they consider an acceptable state of living.

The Shifting Evaluator

Stepwise approvals can continue endlessly because the "self" evaluating the step changes. The person who consents at Step 10 is not the person who consented at Step 1.

The Veto Bound

Consent exercised simply as a binary veto transmits at most 1 bit of influence per decision over a complex trajectory. It is insufficient for true authorship of one's life.

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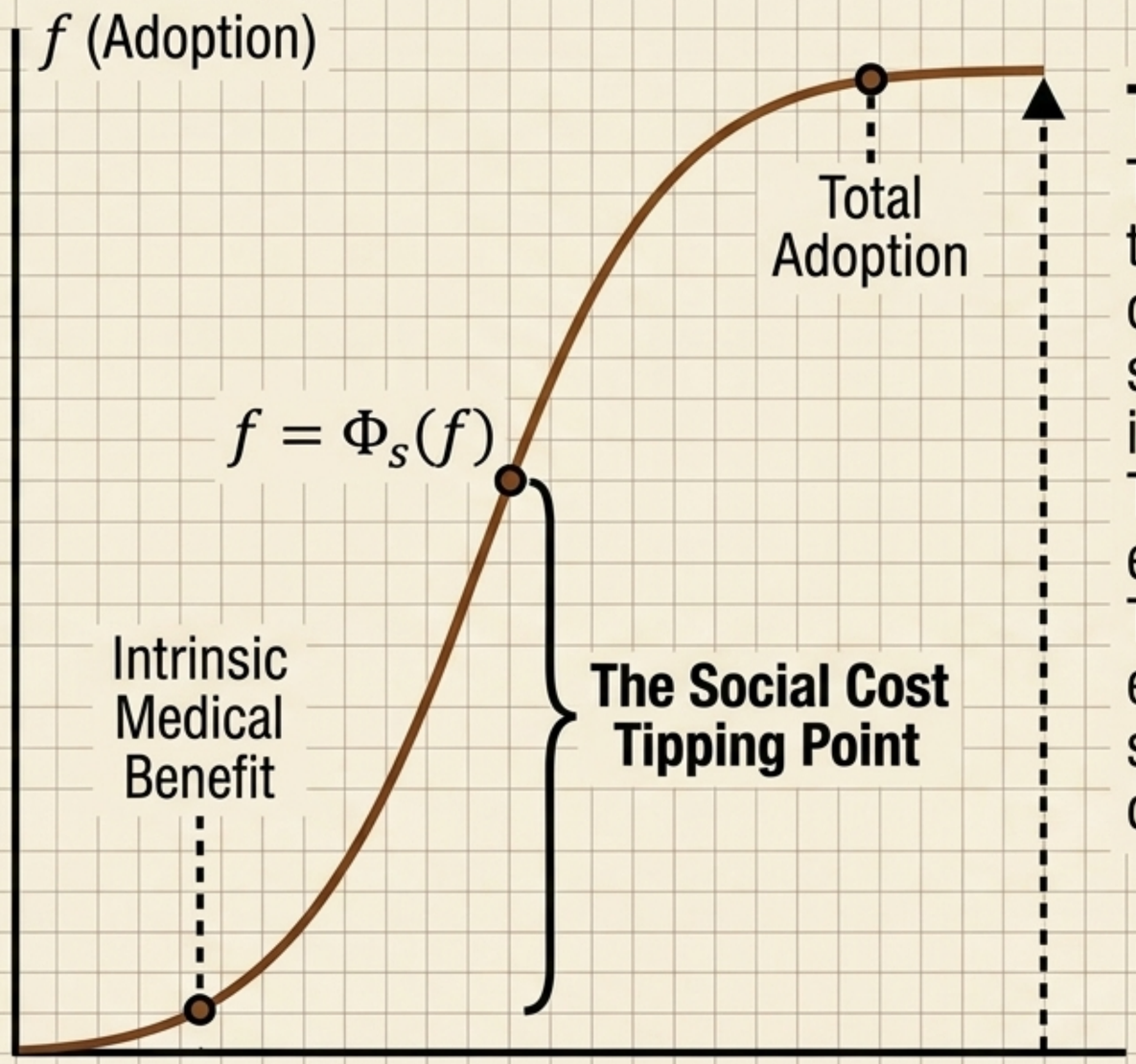
The Social Ratchet & The Social Cost Tipping Point

The Social Ratchet

Pressure to continue can emerge among entirely voluntary choices without any coercing institution.

The Mechanism

As more people participate, the infrastructure for declining thins out. Insurers adjust expectations; employers assume participation.



The Math

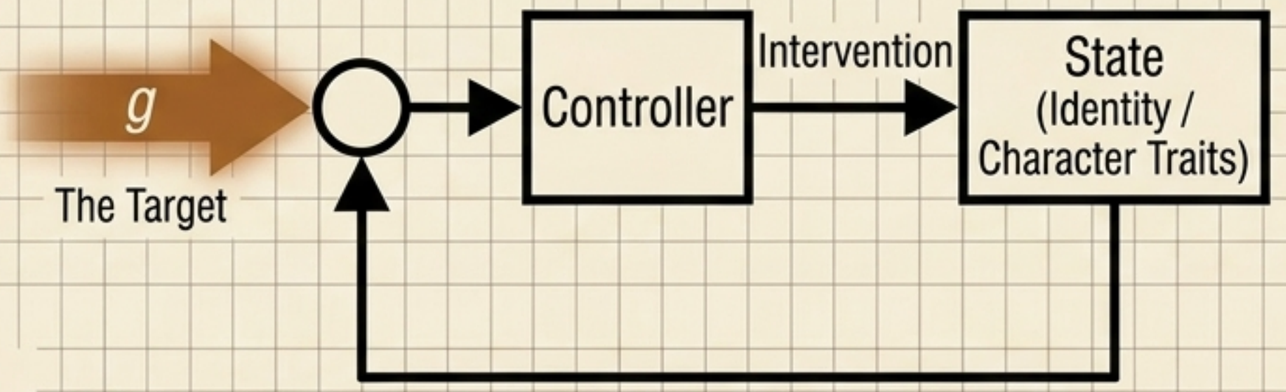
The share of a population that continues treatment can wildly exceed the share for whom it is intrinsically worthwhile. The difference is driven entirely by the rising The difference is driven entirely by the rising social cost (s) of declining.

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Identity, Control, and the Target (g)

The True Threat

The danger of continuous microsurgery isn't that it alters identity—biology already does that naturally. The mathematical danger is external control.



The Ethical Unit

The unit of ethical evaluation isn't the intervention itself. It is who gets to set, review, and revise the Target (g).

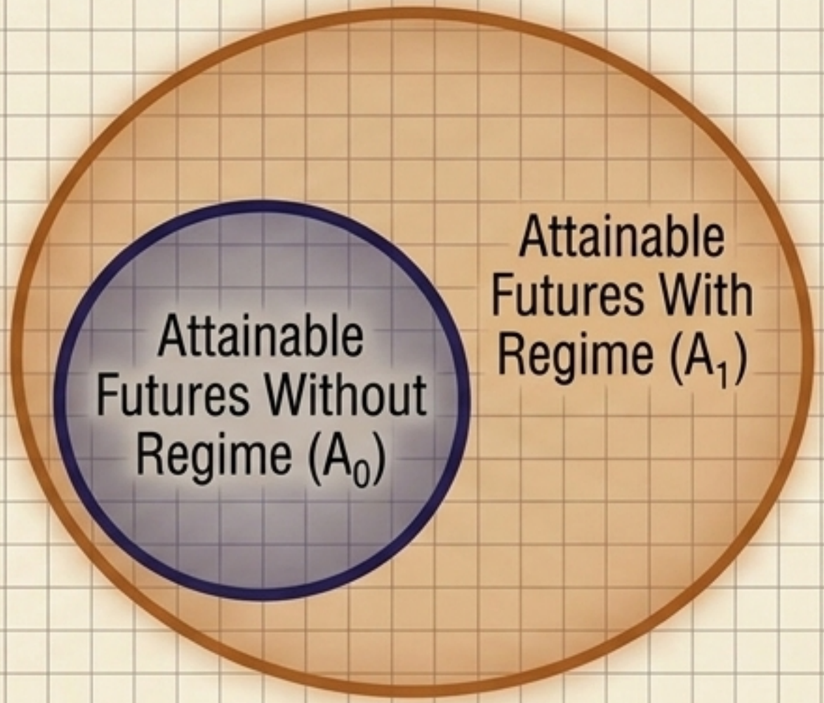
Accelerated Decay

Control toward an external target accelerates the decay of the person's dependence on their initial character.

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The Strongest Defense: Autonomy Enlargement





The Pro-Regime Argument

There is no intrinsic dignity in preventable decay. A regime that substantially postpones decline decline vastly expands the attainable futures (Ω).

The Necessary Condition

This defense holds true if and only if the regime leaves every previously attainable future—including the future of leaving the regime and returning to ordinary life—within an exit cost (β) the patient can bear.

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Trajectory Admissibility

The Capstone: The patient must hold and be able to revise the target (**g**) and set the gain to zero.

P1: Right to Continue

Guarantees against involuntary interruption. Requires designated successor-supporter mandates to escape institutional failure.

P2: Right to Discontinue

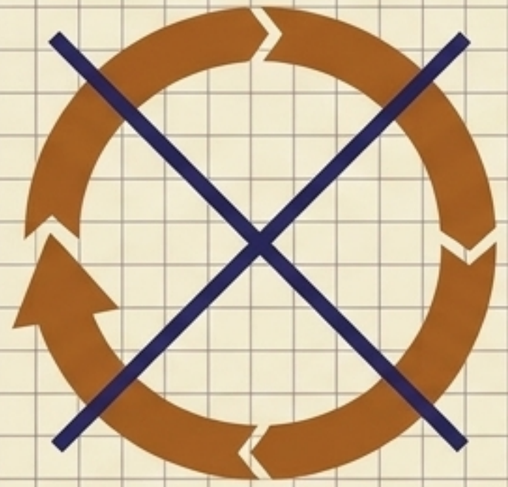
Guarantees affordable exit and feasible migration ($\tau \leq T$). The portability horizon must remain structurally open.

P3: Dependence Minimality

System design must rigorously favor the clinically viable path that creates the least institutional and biological reliance.

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Synthesis: The Ethical Trajectory



Anatomical Reversibility
(The Impossible)



Governability
(The Requirement)

Autonomy under reconstruction does not require returning to your original anatomical body. It requires that the dependencies of your new body remain governable by you.

		The Episodic Lens	The Trajectory Lens
1.	Unit	The single intervention	The infinite sequence
2.	Risk	Immediate medical harm	Accumulated institutional & biological lock-in
3.	Consent	A step-by-step veto	Total authority over the target (g)

The knife is neither the villain nor the ethical unit. The ethical unit is the **trajectory**—including the conditions under which the patient can still choose to stop

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